

- language with language models. In *XAI in Action: Past, Present, and Future Applications*, 2023. URL <https://openreview.net/forum?id=3BX9tM03GT>.
- Slobodkin, A., Goldman, O., Caciularu, A., Dagan, I., and Ravfogel, S. The curious case of hallucinatory (un)answerability: Finding truths in the hidden states of over-confident large language models. In Bouamor, H., Pino, J., and Bali, K. (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 3607–3625, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.220. URL <https://aclanthology.org/2023.emnlp-main.220>.
- Stolfo, A., Belinkov, Y., and Sachan, M. A mechanistic interpretation of arithmetic reasoning in language models using causal mediation analysis. In Bouamor, H., Pino, J., and Bali, K. (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7035–7052, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.435. URL <https://aclanthology.org/2023.emnlp-main.435>.
- Strobelt, H., Gehrmann, S., Pfister, H., and Rush, A. M. Lstmvis: A tool for visual analysis of hidden state dynamics in recurrent neural networks. *IEEE transactions on visualization and computer graphics*, 24(1):667–676, 2017.
- Tenney, I., Das, D., and Pavlick, E. Bert rediscovers the classical nlp pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4593–4601, 2019.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023b.
- Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nevo, D., Singer, Y., and Shieber, S. Investigating gender bias in language models using causal mediation analysis. *Advances in neural information processing systems*, 33: 12388–12401, 2020.
- Vilas, M. G., Schaumlöffel, T., and Roig, G. Analyzing vision transformers for image classification in class embedding space. *arXiv preprint arXiv:2310.18969*, 2023.
- Wallat, J., Singh, J., and Anand, A. BERTnesia: Investigating the capture and forgetting of knowledge in BERT. In Alishahi, A., Belinkov, Y., Chrupala, G., Hupkes, D., Pinter, Y., and Sajjad, H. (eds.), *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 174–183, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.blackboxnlp-1.17. URL <https://aclanthology.org/2020.blackboxnlp-1.17>.
- Wang, B. and Komatsuzaki, A. GPT-J-6B: A 6 Billion Parameter Autoregressive Language Model. <https://github.com/kingoflolz/mesh-transformer-jax>, May 2021.
- Wang, K. R., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations*, 2022.
- Wang, Z., Ku, A., Baldridge, J., Griffiths, T. L., and Kim, B. Gaussian Process Probes (GPP) for uncertainty-aware probing. *arXiv preprint arXiv:2305.18213*, 2023.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35: 24824–24837, 2022.
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., and Narasimhan, K. R. Tree of thoughts: Deliberate problem solving with large language models. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.
- Youssef, P., Koraş, O., Li, M., Schlöterer, J., and Seifert, C. Give me the facts! a survey on factual knowledge probing in pre-trained language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 15588–15605, 2023.
- Zhang, F. and Nanda, N. Towards best practices of activation patching in language models: Metrics and methods. *arXiv preprint arXiv:2309.16042*, 2023.
- Zhong, Z., Wu, Z., Manning, C. D., Potts, C., and Chen, D. Mquake: Assessing knowledge editing in language models via multi-hop questions. *arXiv preprint arXiv:2305.14795*, 2023.
- Zhou, B., Bau, D., Oliva, A., and Torralba, A. Interpreting deep visual representations via network dissection. *IEEE transactions on pattern analysis and machine intelligence*, 41(9):2131–2145, 2018.

A. Detailed Configuration of Prior Methods as Patchscope Instances

In this section, we first reiterate Patchscopes premise and distinguish it from prior work. We then detail Patchscope configurations that replicate prior methods. We highlight that clearly formulating Patchscopes in the form of source and target quintuplets, respectively (S, i, M, l) and (T, i^*, M^*, l^*) , is not in itself a contribution of this work. However, it is what facilitates the main contributions that are 3-fold: 1) The framework ties methods that otherwise might seem unrelated, e.g., logit lens and attention knockout, 2) makes it easy to come up with new configurations that significantly improve them, 3) makes it easy to see the space that prior methods do not cover, and therefore open up new applications altogether.

We focus on a few inspection objectives. Note that some of the baselines considered have not been originally designed to address these objectives, but are the closest solutions in literature. Particularly, we ask, given a (set of) hidden representation(s) detached from their context, how well can we do next-token prediction, feature extraction, and analysis of model’s contextualization process.

A.1. Next-token Prediction

Goal: Estimating output probability distribution p^L given a hidden representation h^l , particularly to inspect when the model has enough information to make the final prediction.

Prior Work: Many recent methods to estimate output probability distribution given an inner representation use projection onto the vocabulary space (e.g., [nostalgebraist, 2020](#); [Din et al., 2023](#); [Belrose et al., 2023](#)). Considering (S, i, M, l) and (T, i^*, M^*, l^*) formulation above, all these methods can be seen as Patchscopes where $\mathbf{M} = \mathbf{M}^*$, $\mathbf{l}^* = \mathbf{L}$, where $\mathbf{L}$ is the number of layers.

Distinguishing Patchscope & its Improvements: We propose the token identity Patchscope that unlike ([Belrose et al., 2023](#); [Din et al., 2023](#)) does not require a learned mapping, yet is more successful. Note that in this task, setting $\mathbf{l}^* = \mathbf{L}$ which all the prior methods discussed above have in common, means that the choice of $\mathbf{T}$ does not matter. However, in our proposed Patchscope, we show that setting $\mathbf{l}^* = \mathbf{l}$ where $\mathbf{l}$ is arbitrary layer, and using a few-shot token identity prompt $\mathbf{T}$ that encourages repetition results in significant improvements, and by using the target model’s computation from layer $\mathbf{l}^*$ onward, there is no need for training a separate mapping function.

A.2. Feature Extraction

Goal: Extracting specific attributes of the subject given its hidden representation h^l (e.g., extracting “largest city of” from the representation of “States” in “United States”).

Prior Work: Classification probes ([Alain & Bengio, 2017](#)) are the most commonly used methods for this purpose. They work by training a classifier on a dataset of prompts and their corresponding labels (e.g., countries and their corresponding largest cities).

Distinguishing Patchscope & its Improvements: We show an intuitive Patchscope (i.e., $T = \text{“The largest city in x”}$) significantly outperforms probing across various tasks and layers. Note that this Patchscope 1) does not require supervised training, 2) is not limited by a predefined set of labels, and 3) is more accurate than a linear probe.

A.3. Analysis of Model’s Contextualization Process

Goal: Identifying how entity tokens are processed, and at which layer the entity is fully resolved.

Prior Work: It is hard to answer this question with existing methods. First, methods that use vocabulary projection are focused on output prediction (rather than input processing) and fail particularly in early layers, precisely where entity resolution happens. Second, probing requires a predefined set of classes, which is not expressive enough to show the gradual entity resolution process, and nontrivial. For example, consider “Alexander the grea” input. We found that “Great Britain” and “Great depression” arose in the gradual resolution process. However, these are non-trivial choices a priori. We can train a multi-class classifier to choose one out of all possible entities (or binary classifiers for every possible entity), which is unlikely to work given the huge space of possible entities. Lastly, most prior work using activation patching aims to answer a different question: whether a certain representation plays a key role in the model’s forward computation and the

final prediction (e.g., Meng et al., 2022a; Geva et al., 2023). However, they can be viewed as Patchscope instances, and provide indirect signals about the entity resolution process, namely, showing subject information accrues in the last token after a few layers.

Distinguishing Patchscope & its Improvements: To the best of our knowledge, there are not any methods that verbalize the gradual entity resolution process in coherent text, and the proposed Patchscope is the first to do that.

In summary, as discussed in 3.2, many prior methods can be seen as Patchscopes. Tab. 4 details the configurations of Patchscopes that yield various methods introduced in prior work.

Table 4. Patchscopes is a novel framework for inspection of hidden representations in language models. Many prior inspection methods with various objectives can be viewed as Patchscopes, as detailed in the “Configuration” column (see notation description in §3). The rows highlighted in green show our new configurations that overcome several limitations of prior methods through more expressive inspection that is training-data free and is more robust across layers. Additionally, the generality of this framework enables novel inspection possibilities that were unexplored before. When the target prompt (T) is not specified, it means that the output would be invariant to the choice of T . When not specified, $f \leftarrow \mathbb{I}$ and $\mathcal{M}^* = \mathcal{M}$.

Inspection Objective		Expressive	Training Data Free	Robust Across Layers	Configuration
Inspecting Output Distribution	Few-shot token identity Patchscope (§4.1)	✓✓	✓	✓✓	$\ell^* \leftarrow \ell$, $T \leftarrow \text{"tok}_1 \rightarrow \text{tok}_1; \text{tok}_2 \rightarrow \text{tok}_2; \dots; \text{tok}_k \text{"}$
	Logit Lens (nostalgebraist, 2020), Embedding Space Analysis (Dar et al., 2023)	✓	✓	✗	$\ell^* \leftarrow L^*$
	Tuned Lens (Belrose et al., 2023)	✓	For learning mappings	✓	$\ell^* \leftarrow L^*$, $f \leftarrow \text{Affine}$
	Future Lens (Pal et al., 2023)	✓	For learning mappings	✓✓	$\ell^* \leftarrow \ell$, $f \leftarrow \text{Linear}$, $T \leftarrow \text{Fixed or learned soft prompt}$
Feature Extraction	Zero-shot feature extraction Patchscope (§4.2)	✓✓	✓	✓✓	$\ell^* \leftarrow j' \in [1, \dots, L^*]$, $i^* \leftarrow m$, $T \leftarrow \text{relation verbalization followed by } x$
	LRE Attribute Lens (Hernandez et al., 2023b)	✓	For linear relation approx.	✓✓	$\ell^* \leftarrow L^*$, $f \leftarrow \text{Linear with additional variables}$, $T \leftarrow S$
	Probing (e.g., Belinkov & Glass, 2019; Belinkov, 2022; Alain & Bengio, 2017; Wang et al., 2023)	✗	For training probe	✓	N/A
Entity Resolution	Entity description Patchscope (§4.3)	✓✓	✓	✓✓	$\ell^* \leftarrow \ell$, $i^* \leftarrow m$, $T \leftarrow \text{"subject}_1: \text{description}_1, \dots, \text{subject}_k: \text{description}_k, x \text{"}$
	X-model entity description Patchscope (§4.4)	✓✓✓	For learning mappings	✓✓	$\mathcal{M}^* \leftarrow \text{a larger variant of } \mathcal{M}$, $\ell^* \leftarrow \ell$, $i^* \leftarrow m$, $T \leftarrow \text{"subject}_1: \text{description}_1, \dots, \text{subject}_k: \text{description}_k, x \text{"}$
	Causal Tracing (Meng et al., 2022a)	✗	✓	✓✓	$\ell^* \leftarrow \ell$, $T \leftarrow S + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma)$
	Attention Knockout (Wang et al., 2022; Conmy et al., 2023; Geva et al., 2023)	✗	✓	✓✓	$\ell^* \leftarrow \text{Multiple}$, $f \leftarrow 0$, $T \leftarrow S$
Inspection Application	Early Exiting, e.g., Linear Shortcuts (Din et al., 2023)	✓	For learning mappings	✓	$\ell^* \leftarrow L^*$, $f \leftarrow \text{Affine}$
	Caption Generation, e.g., Linear Mapping (Merullo et al., 2022)	✓	For learning mappings	✓	$\mathcal{M}^* \leftarrow \text{A language model of choice}$, $\ell^* \leftarrow L^*$, $f \leftarrow \text{Affine}$

B. Next-Token Prediction Additional Details and Experimental Results

In this section, we provide more information about the models, data, and additional target prompts for few-shot token identity Patchscope configurations. It is worth highlighting that after patching, we continue computations from the target layer onward. Therefore, since in this Patchscope, source layer and target layer are the same, there are going to be more computations compared to baseline methods like LogitLens (nostalgebraist, 2020; Geva et al., 2022a) and Tuned Lens (Belrose et al., 2023; Din et al., 2023) where the target layer is fixed at the final layer.

B.1. Models

We use LLaMA2 (13B)⁴ (Touvron et al., 2023b), Vicuna (13B)⁵ (Chiang et al., 2023), GPT-J (6B) (Wang & Komatsuzaki, 2021)⁶, and Pythia (12B)⁷ (Biderman et al., 2023). LLaMA2 was pre-trained on 2T tokens from a mix of publicly available data. Vicuna is a LLaMA1 (Touvron et al., 2023a) model that was pre-trained on 1T tokens and fine-tuned on 70K

⁴LLaMA 2 is licensed under the LLAMA 2 Community License, Copyright © Meta Platforms, Inc. All Rights Reserved.

⁵Vicuna is subject to the LLAMA 2 Community License, terms of use of the data generated by OpenAI, and privacy practices of ShareGPT. The code is released under the Apache License 2.0.

⁶GPT-J-6B weights are licensed under version 2.0 of the Apache License.

⁷Pythia is released under Apache License 2.0, Copyright © 2023 EleutherAI.